AI 101 FOR MECHANICAL ENGINEERS

MODULE 15 / LESSON 2

Verification and validation are different checks



Undergraduate course

Lesson outcomes

Distinguish implementation verification from real-system validation.

Record provenance and reproduce a stated procedure.

Verification and validation

Verification asks whether the intended equations or algorithm were implemented correctly.

Check units, known examples, bounds, limiting cases, sign conventions and reproducibility.

Validation asks whether the model adequately represents the real system for the intended purpose, using appropriate independent physical evidence.

A numerical check and a physical comparison

For a power model, confirm that ΔpQ has watts as units and that a known test computes correctly.

Then compare predictions with calibrated measurements across relevant conditions.

A simulator matching itself in synthetic examples is useful verification of a workflow, not validation of real hardware.

Provenance and reproducibility

Keep provenance: acquisition details, transformations, split IDs, code and versions.

Reproducibility means another person can regenerate the stated procedure. Repeated wrong results are still wrong.

Explain limitations proportional to evidence.

Lesson vocabulary

Verification: Checking correct implementation of the intended method. Example: Test units and a known calculation.

Validation: Checking suitability against real-system evidence for the intended use. Example: Compare with calibrated experiments.

Provenance: Records of where data and results came from. Example: Keep source, units, transformations and versions.

Worked exercise

Is matching a synthetic simulator enough to validate predictions for real equipment?

Pause and write an answer before moving to the explanation.

Exercise explanation

No. Agreement on synthetic data can verify a procedure, but real-equipment validation needs independent physical evidence for the intended use.

Verification, validation and a changed sensor

A notebook passes a known arithmetic example and fits synthetic pump data. After a new sensor is installed, its readings shift. What has been verified, and what further evidence is needed?

Attempt the problem before reading the worked explanation.

Worked explanation

The known case provides evidence about implementation correctness. The synthetic study checks a declared teaching workflow.

Neither establishes that the model represents real equipment across its intended operating conditions. Independent calibrated measurements are needed for that validation.

The sensor change may create distribution shift. Investigate calibration, units and acquisition before interpreting the model output.

Define a named reviewer, a deferral rule and criteria for new validation or retraining. Record the model and sensor versions.

Concept reference table 1

Check	Question	Evidence
Verification	Is the implementation correct?	Known cases and units
Validation	Does it represent the intended real system?	Independent physical comparison
Reproducibility	Can the stated procedure be regenerated?	Inputs, code and environment

Teaching example. Read the table alongside the worked explanation.

Concept reference table 2

Check	Question	Evidence
Provenance	Where did evidence originate?	Acquisition and transformation record

Teaching example. Read the table alongside the worked explanation.

Knowledge check 1

Testing a known calculation checks…

A. Implementation verification

B. All real-system validity

C. AGI

D. Only branding

Knowledge check 2

Validation for real machinery needs…

A. Relevant independent real-system evidence

B. Only synthetic self-comparison

C. Only a random seed

D. A fluent answer

Knowledge-check explanations

Check 1: A. A known case helps verify implementation.

Check 2: A. Validate for the intended use with suitable measurements.

Lesson recap

Distinguish implementation verification from real-system validation.

Record provenance and reproduce a stated procedure.

Apply the ideas to this question: Is matching a synthetic simulator enough to validate predictions for real equipment?

Sources and further reading

Primary source: AI 101 for Mechanical Engineers, Module 15, Lesson 2: Verification and validation are different checks.

Worked cases and numerical examples are teaching material. Synthetic evidence does not validate real equipment.

Google Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

scikit-learn User Guide
https://scikit-learn.org/stable/user_guide.html

Technical sources (numbers from the course reference list):
Oberkampf and Roy [13], verification and validation; Pineau et al. [36], reproducibility; Gebru et al. [38], data documentation.